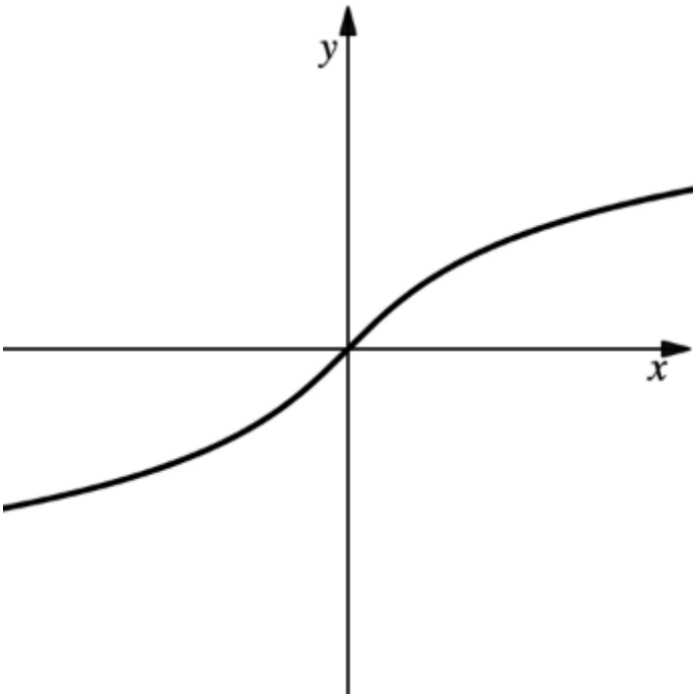
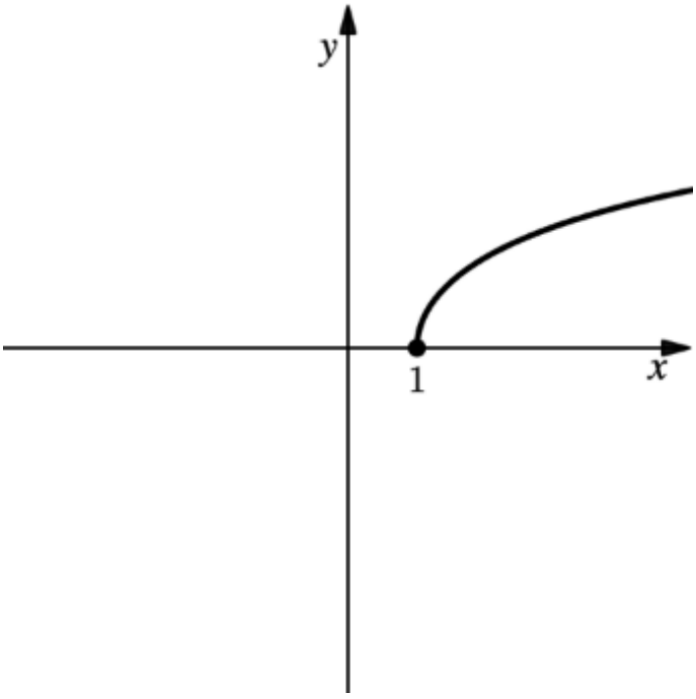
2.16 Hyperbolic Functions: Inverse Functions

Worked Solutions

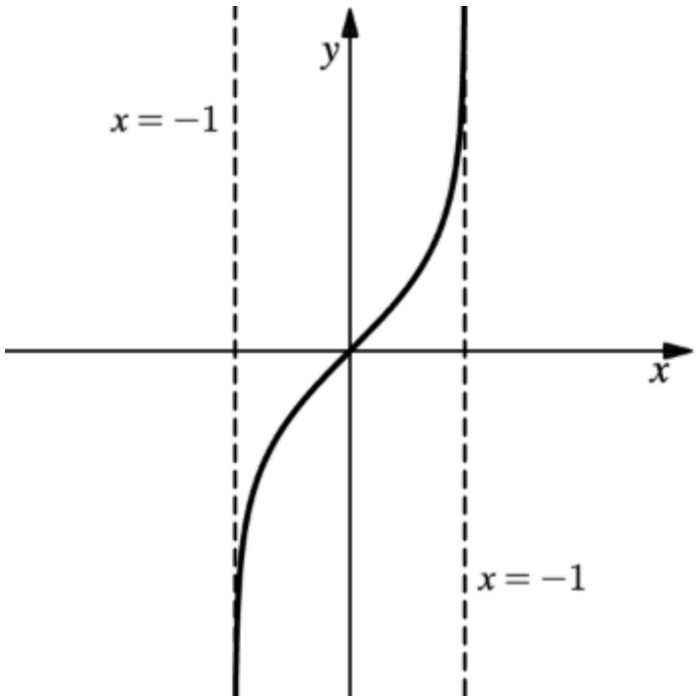
1 a)



1 b)



1 c)



2)

$$\begin{aligned}
\text{Let } y = \sinh x &= \frac{e^x - e^{-x}}{2}. \\
\text{Swap } x\text{s and } y\text{s: } x &= \frac{e^y - e^{-y}}{2} \\
\Rightarrow x &= \frac{e^{2y} - 1}{2e^y} \\
\Rightarrow 2xe^y &= e^{2y} - 1 \\
\Rightarrow e^{2y} - 2xe^y - 1 &= 0 \\
\Rightarrow (e^y - x)^2 - x^2 - 1 &= 0 \\
\Rightarrow (e^y - x)^2 &= x^2 + 1 \\
\Rightarrow e^y - x &= \pm\sqrt{x^2 + 1} \\
\Rightarrow e^y &= x \pm \sqrt{x^2 + 1}.
\end{aligned}$$

We take the positive square root only, as $e^y > 0$ ($x - \sqrt{x^2 + 1} < 0$ for all real values of x).

$$\begin{aligned}
\Rightarrow y &= \ln(x + \sqrt{x^2 + 1}) \\
\therefore \operatorname{arsinh} x &= \ln(x + \sqrt{x^2 + 1}).
\end{aligned}$$

3)

$$\text{Let } \frac{y}{k} = \sinh x = \frac{e^x - e^{-x}}{2}.$$

$$\begin{aligned}
\text{Swap } x\text{s and } y\text{s: } \frac{x}{k} &= \frac{e^y - e^{-y}}{2} \\
\Rightarrow \frac{x}{k} &= \frac{e^{2y} - 1}{2e^y} \\
\Rightarrow \frac{2xe^y}{k} &= e^{2y} - 1 \\
\Rightarrow e^{2y} - \frac{2xe^y}{k} - 1 &= 0 \\
\Rightarrow \left(e^y - \frac{x}{k}\right)^2 - \frac{x^2}{k^2} - 1 &= 0 \\
\Rightarrow \left(e^y - \frac{x}{k}\right)^2 &= \frac{x^2}{k^2} + 1 \\
\Rightarrow e^y - \frac{x}{k} &= \pm\sqrt{\frac{x^2}{k^2} + 1} \\
\Rightarrow e^y &= \frac{x}{k} \pm \sqrt{\frac{x^2}{k^2} + 1}.
\end{aligned}$$

We take the positive square root only, as $e^y > 0$ ($\frac{x}{k} - \sqrt{\frac{x^2}{k^2} + 1} < 0$ for all real values of x and for $k > 0$).

$$\begin{aligned}
\Rightarrow y &= \ln\left(\frac{x}{k} + \sqrt{\frac{x^2}{k^2} + 1}\right) \\
&= \ln\left(\frac{x}{k} + \sqrt{\frac{1}{k^2}(x^2 + k^2)}\right) \\
&= \ln\left(\frac{x}{k} + \frac{1}{k}\sqrt{x^2 + k^2}\right) \\
&= \ln\left(\frac{x + \sqrt{x^2 + k^2}}{k}\right) \\
\therefore \operatorname{arsinh}\frac{x}{k} &= \ln\left(\frac{x + \sqrt{x^2 + k^2}}{k}\right).
\end{aligned}$$

4)

$$\begin{aligned}
\text{Let } y = \cosh x &= \frac{e^x + e^{-x}}{2}. \\
\text{Swap } x\text{s and } y\text{s: } x &= \frac{e^y + e^{-y}}{2} \\
\Rightarrow x &= \frac{e^{2y} + 1}{2e^y} \\
\Rightarrow 2xe^y &= e^{2y} + 1 \\
\Rightarrow e^{2y} - 2xe^y + 1 &= 0 \\
\Rightarrow (e^y - x)^2 - x^2 + 1 &= 0 \\
\Rightarrow (e^y - x)^2 &= x^2 - 1 \\
\Rightarrow e^y - x &= \pm\sqrt{x^2 - 1} \\
\Rightarrow e^y &= x \pm \sqrt{x^2 - 1}.
\end{aligned}$$

By definition, for $y = \operatorname{arcosh} x$ this means that $y \geq 0$. Therefore, $e^y \geq 1$, so the correct root must be at least 1. For $x \geq 1$, $x + \sqrt{x^2 - 1} \geq x \geq 1$, so this works. For $x > 1$, $\sqrt{x^2 - 1} < x$ and $x - \sqrt{x^2 - 1} < 1$, hence the negative root cannot equal e^y.

$$\begin{aligned} \Rightarrow y &= \ln(x + \sqrt{x^2 - 1}) \\ \therefore \operatorname{arcosh} x &= \ln(x + \sqrt{x^2 - 1}). \end{aligned}$$

5)

$$\begin{aligned} \text{Let } \frac{y}{k} = \cosh x &= \frac{e^x + e^{-x}}{2}. \\ \text{Swap } x\text{s and } y\text{s: } \frac{x}{k} &= \frac{e^y + e^{-y}}{2} \\ \Rightarrow \frac{x}{k} &= \frac{e^{2y} + 1}{2e^y} \\ \Rightarrow \frac{2xe^y}{k} &= e^{2y} + 1 \\ \Rightarrow e^{2y} - \frac{2xe^y}{k} + 1 &= 0 \\ \Rightarrow \left(e^y - \frac{x}{k}\right)^2 - \frac{x^2}{k^2} + 1 &= 0 \\ \Rightarrow \left(e^y - \frac{x}{k}\right)^2 &= \frac{x^2}{k^2} - 1 \\ \Rightarrow e^y - \frac{x}{k} &= \pm\sqrt{\frac{x^2}{k^2} - 1} \\ \Rightarrow e^y &= \frac{x}{k} \pm \sqrt{\frac{x^2}{k^2} - 1}. \end{aligned}$$

By definition, for $y = \operatorname{arcosh} \frac{x}{k}$ this means that $y \geq 0$. Therefore $e^y \geq 1$, so the correct root must be at least 1. For $\frac{x}{k} \geq 1$, $\frac{x}{k} + \sqrt{\frac{x^2}{k^2} - 1} \geq \frac{x}{k} \geq 1$, so this works. For $\frac{x}{k} > 1$, $\sqrt{\frac{x^2}{k^2} - 1} < \frac{x}{k}$ and $\frac{x}{k} - \sqrt{\frac{x^2}{k^2} - 1} < 1$, hence the negative root cannot equal e^y.

$$\begin{aligned} \Rightarrow y &= \ln\left(\frac{x}{k} + \sqrt{\frac{x^2}{k^2} - 1}\right) \\ &= \ln\left(\frac{x}{k} + \sqrt{\frac{1}{k^2}(x^2 - k^2)}\right) \\ &= \ln\left(\frac{x}{k} + \frac{1}{k}\sqrt{x^2 - k^2}\right) \\ &= \ln\left(\frac{x + \sqrt{x^2 - k^2}}{k}\right) \\ \therefore \operatorname{arcosh} \frac{x}{k} &= \ln\left(\frac{x + \sqrt{x^2 - k^2}}{k}\right). \end{aligned}$$

6)

$$\text{Let } y = \tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}.$$

$$
\begin{aligned}
\text{Swap } x\text{s and } y\text{s: } x &= \frac{e^y - e^{-y}}{e^y + e^{-y}} \\
\Rightarrow x &= \frac{e^{2y} - 1}{e^{2y} + 1} \\
\Rightarrow xe^{2y} + x &= e^{2y} - 1 \\
\Rightarrow 1 + x &= e^{2y} - xe^{2y} \\
\Rightarrow 1 + x &= e^{2y}(1 - x) \\
\Rightarrow e^{2y} &= \frac{1 + x}{1 - x} \\
\Rightarrow 2y &= \ln\left(\frac{1 + x}{1 - x}\right) \\
y &= \frac{1}{2}\ln\left(\frac{1 + x}{1 - x}\right) \\
\therefore \operatorname{artanh} x &= \frac{1}{2}\ln\left(\frac{1 + x}{1 - x}\right).
\end{aligned}
$$

7)

$$
\begin{aligned}
\text{Let } \frac{y}{k} = \tanh x &= \frac{e^x - e^{-x}}{e^x + e^{-x}}. \\
\text{Swap } x\text{s and } y\text{s: } \frac{x}{k} &= \frac{e^y - e^{-y}}{e^y + e^{-y}} \\
\Rightarrow \frac{x}{k} &= \frac{e^{2y} - 1}{e^{2y} + 1} \\
\Rightarrow \frac{xe^{2y}}{k} + \frac{x}{k} &= e^{2y} - 1 \\
\Rightarrow 1 + \frac{x}{k} &= e^{2y} - \frac{xe^{2y}}{k} \\
\Rightarrow 1 + \frac{x}{k} &= e^{2y}\left(1 - \frac{x}{k}\right) \\
\Rightarrow e^{2y} &= \frac{1 + x/k}{1 - x/k} \\
&= \frac{k + x}{k - x} \\
\Rightarrow 2y &= \ln\left(\frac{k + x}{k - x}\right) \\
y &= \frac{1}{2}\ln\left(\frac{k + x}{k - x}\right) \\
\therefore \operatorname{artanh}\frac{x}{k} &= \frac{1}{2}\ln\left(\frac{k + x}{k - x}\right).
\end{aligned}
$$

8)

$$
\begin{aligned}
\operatorname{arcosh} x &= \ln 10 \\
\Rightarrow \ln(x + \sqrt{x^2 - 1}) &= \ln 10 \\
\Rightarrow x + \sqrt{x^2 - 1} &= 10 \\
\Rightarrow \sqrt{x^2 - 1} &= 10 - x
\end{aligned}
$$

$$\begin{aligned}
\Rightarrow x^2 - 1 &= (10 - x)^2 \\
\Rightarrow x^2 - 1 &= 100 - 20x + x^2 \\
\Rightarrow -1 &= 100 - 20x \\
\Rightarrow 20x &= 101 \\
\Rightarrow x &= \frac{101}{20}.
\end{aligned}$$

Alternatively,

$$\begin{aligned}
\operatorname{arcosh} x &= \ln 10 \\
\Rightarrow \cosh(\operatorname{arcosh} x) &= \cosh(\ln 10) \\
x &= \frac{e^{\ln 10} + e^{-\ln 10}}{2} = \frac{10 + 1/10}{2} = \frac{101}{20}.
\end{aligned}$$

9)

$$\begin{aligned}
\operatorname{arsinh} x &= -\ln 6 \\
\Rightarrow \ln(x + \sqrt{x^2 + 1}) &= \ln 6^{-1} = \ln \frac{1}{6} \\
\Rightarrow x + \sqrt{x^2 + 1} &= \frac{1}{6} \\
\Rightarrow 6x + 6\sqrt{x^2 + 1} &= 1 \\
\Rightarrow 6\sqrt{x^2 + 1} &= 1 - 6x \\
\Rightarrow 36(x^2 + 1) &= (1 - 6x)^2 \\
\Rightarrow 36x^2 + 36 &= 1 - 12x + 36x^2 \\
\Rightarrow 36 &= 1 - 12x \\
\Rightarrow 12x &= -35 \\
\Rightarrow x &= -\frac{35}{12}.
\end{aligned}$$

Alternatively,

$$\begin{aligned}
\operatorname{arsinh} x &= -\ln 6 \\
\Rightarrow \sinh(\operatorname{arsinh} x) &= \sinh(-\ln 6) \\
x &= \frac{e^{-\ln 6} - e^{\ln 6}}{2} = \frac{1/6 - 6}{2} = -\frac{35}{12}.
\end{aligned}$$

10)

$$\begin{aligned}
\operatorname{artanh} \frac{x}{6} &= -\ln 2 \\
\Rightarrow \frac{1}{2} \ln \left(\frac{6 + x}{6 - x} \right) &= \ln 2^{-1} \\
\Rightarrow \ln \left(\frac{6 + x}{6 - x} \right) &= 2 \ln \frac{1}{2} \\
\Rightarrow \ln \left(\frac{6 + x}{6 - x} \right) &= \ln \left(\frac{1}{2} \right)^2 = \ln \frac{1}{4}
\end{aligned}$$

$$\begin{aligned}
\Rightarrow \frac{6+x}{6-x} &= \frac{1}{4} \\
\Rightarrow 24+4x &= 6-x \\
\Rightarrow 5x &= -18 \\
\Rightarrow x &= -\frac{18}{5}.
\end{aligned}$$

Alternatively,

$$\begin{aligned}
\operatorname{artanh}\frac{x}{6} &= -\ln 2 \\
\tanh\left(\operatorname{artanh}\frac{x}{6}\right) &= \tanh(-\ln 2) \\
\frac{x}{6} &= \frac{e^{-\ln 2}-e^{\ln 2}}{e^{-\ln 2}+e^{\ln 2}} = \frac{1/2-2}{1/2+2} = -\frac{3}{5} \\
x &= -\frac{18}{5}.
\end{aligned}$$

11 a)

$$\begin{aligned}
\operatorname{artanh} x &= \ln(\sqrt{x+2}) \\
\Rightarrow \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right) &= \ln(\sqrt{x+2}) \\
\Rightarrow \ln\left(\frac{1+x}{1-x}\right) &= 2\ln(\sqrt{x+2}) \\
\Rightarrow \ln\left(\frac{1+x}{1-x}\right) &= \ln((\sqrt{x+2})^2) \\
\Rightarrow \frac{1+x}{1-x} &= x+2 \\
\Rightarrow 1+x &= (x+2)(1-x) \\
\Rightarrow 1+x &= -x^2-x+2 \\
\Rightarrow x^2+2x-1 &= 0 \\
\Rightarrow x &= -1\pm\sqrt{2}.
\end{aligned}$$

$x \neq -1-\sqrt{2}$ as $-1 < x < 1$. So $x = -1+\sqrt{2}$ is the only real solution. Alternatively,

$$\begin{aligned}
\operatorname{artanh} x &= \ln(\sqrt{x+2}) \\
\tan(\operatorname{artanh} x) &= \tan(\ln(\sqrt{x+2})) \\
x &= \frac{e^{\ln(\sqrt{x+2})}-e^{-\ln(\sqrt{x+2})}}{e^{\ln(\sqrt{x+2})}+e^{-\ln(\sqrt{x+2})}} \\
\Rightarrow x &= \frac{\sqrt{x+2}-1/\sqrt{x+2}}{\sqrt{x+2}+1/\sqrt{x+2}} = \frac{x+2-1}{x+2+1} \\
&= \frac{x+1}{x+3} \\
\Rightarrow x(x+3) &= x+1 \\
\Rightarrow x^2+3x-x-1 &= 0 \\
\Rightarrow x^2+2x-1 &= 0 \\
\Rightarrow x &= -1\pm\sqrt{2}.
\end{aligned}$$

$x \neq -1-\sqrt{2}$ as $-1 < x < 1$. So $x = -1+\sqrt{2}$ is the only real solution.

11 b)

$$\begin{aligned}
\operatorname{artanh}\frac{x}{3} &= \ln(\sqrt{3-2x}) \\
\Rightarrow \frac{1}{2}\ln\left(\frac{3+x}{3-x}\right) &= \ln(\sqrt{3-2x}) \\
\Rightarrow \ln\left(\frac{3+x}{3-x}\right) &= \ln((\sqrt{3-2x})^2) \\
\Rightarrow \frac{3+x}{3-x} &= 3-2x \\
\Rightarrow 3+x &= (3-2x)(3-x) \\
\Rightarrow 3+x &= 2x^2-9x+9 \\
\Rightarrow 0 &= 2x^2-10x+6 \\
\Rightarrow x &= \frac{5}{2} \pm \frac{\sqrt{13}}{2}.
\end{aligned}$$

$x \neq \frac{5}{2} + \frac{\sqrt{13}}{2}$ as artanh $\frac{x}{3}$ is only valid for $-3 < x < 3$. So $x = \frac{5}{2} - \frac{\sqrt{13}}{2}$ is the only real solution. Alternatively,

$$\begin{aligned}
\operatorname{artanh}\frac{x}{3} &= \ln(\sqrt{3-2x}) \\
\tan\left(\operatorname{artanh}\frac{x}{3}\right) &= \tan(\ln(\sqrt{3-2x})) \\
\frac{x}{3} &= \frac{\sqrt{3-2x}-1/\sqrt{3-2x}}{\sqrt{3-2x}+1/\sqrt{3-2x}} = \frac{3-2x-1}{3-2x+1} \\
&= \frac{2-2x}{4-2x} \\
\Rightarrow x(4-2x) &= 3(2-2x) \\
\Rightarrow 4x-2x^2 &= 6-6x \\
\Rightarrow 0 &= 2x^2-10x+6 \\
\Rightarrow x &= \frac{5}{2} \pm \frac{\sqrt{13}}{2}.
\end{aligned}$$

$x \neq \frac{5}{2} + \frac{\sqrt{13}}{2}$ as artanh $\frac{x}{3}$ is only valid for $-3 < x < 3$. So $x = \frac{5}{2} - \frac{\sqrt{13}}{2}$ is the only real solution.

11 c)

$$\begin{aligned}
\operatorname{artanh} 5x &= \ln(\sqrt{2+5x}) \\
\Rightarrow \frac{1}{2}\ln\left(\frac{1+5x}{1-5x}\right) &= \ln(\sqrt{2+5x}) \\
\Rightarrow \ln\left(\frac{1+5x}{1-5x}\right) &= \ln((\sqrt{2+5x})^2) \\
\Rightarrow \frac{1+5x}{1-5x} &= 2+5x \\
\Rightarrow 1+5x &= (2+5x)(1-5x) \\
\Rightarrow 1+5x &= -25x^2-5x+2 \\
\Rightarrow 25x^2+10x-1 &= 0 \\
\Rightarrow x &= -\frac{1}{5} \pm \frac{\sqrt{2}}{5}.
\end{aligned}$$

$x \neq -\frac{1}{5} - \frac{\sqrt{2}}{5}$ as $\operatorname{artanh} 5x$ is only valid for $-\frac{1}{5} < x < \frac{1}{5}$. So $x = -\frac{1}{5} + \frac{\sqrt{2}}{5}$ is the only real solution. Alternatively,

$$\begin{aligned}
\operatorname{artanh} 5x &= \ln(\sqrt{2+5x}) \\
\tan(\operatorname{artanh} 5x) &= \tan(\ln(\sqrt{2+5x})) \\
5x &= \frac{\sqrt{2+5x} - 1/\sqrt{2+5x}}{\sqrt{2+5x} + 1/\sqrt{2+5x}} = \frac{2+5x-1}{2+5x+1} \\
&= \frac{1+5x}{3+5x} \\
\Rightarrow 5x(3+5x) &= 1+5x \\
\Rightarrow 15x + 25x^2 &= 1+5x \\
\Rightarrow 25x^2 + 10x - 1 &= 0 \\
\Rightarrow x &= -\frac{1}{5} \pm \frac{\sqrt{2}}{5}.
\end{aligned}$$

$x \neq -\frac{1}{5} - \frac{\sqrt{2}}{5}$ as $\operatorname{artanh} 5x$ is only valid for $-\frac{1}{5} < x < \frac{1}{5}$. So $x = -\frac{1}{5} + \frac{\sqrt{2}}{5}$ is the only real solution.

11 d)

$$\begin{aligned}
4x+1 &= \tanh(\ln(\sqrt{5-2x})) \\
\Rightarrow \operatorname{artanh}(4x+1) &= \ln(\sqrt{5-2x}) \\
\Rightarrow \frac{1}{2}\ln\left(\frac{1+(4x+1)}{1-(4x+1)}\right) &= \ln(\sqrt{5-2x}) \\
\Rightarrow \ln\left(\frac{2+4x}{-4x}\right) &= 2\ln(\sqrt{5-2x}) \\
\Rightarrow \ln\left(\frac{2+4x}{-4x}\right) &= \ln((\sqrt{5-2x})^2) \\
\Rightarrow \frac{2+4x}{-4x} &= 5-2x \\
\Rightarrow 2+4x &= (5-2x)(-4x) \\
\Rightarrow 2+4x &= 8x^2 - 20x \\
\Rightarrow 0 &= 8x^2 - 24x - 2 \\
\Rightarrow x &= \frac{3}{2} \pm \frac{\sqrt{10}}{2}.
\end{aligned}$$

$x \neq \frac{3}{2} + \frac{\sqrt{10}}{2}$ as $\operatorname{artanh}(4x+1)$ is only valid for $-1 < 4x+1 < 1 \Rightarrow -\frac{1}{2} < x < 0$. So $x = \frac{3}{2} - \frac{\sqrt{10}}{2}$ is the only real solution. Alternatively,

$$\begin{aligned}
4x+1 &= \tanh(\ln(\sqrt{5-2x})) \\
4x+1 &= \frac{\sqrt{5-2x} - 1/\sqrt{5-2x}}{\sqrt{5-2x} + 1/\sqrt{5-2x}} \\
\Rightarrow 4x+1 &= \frac{5-2x-1}{5-2x+1} = \frac{4-2x}{6-2x} \\
\Rightarrow (4x+1)(6-2x) &= 4-2x \\
\Rightarrow 24x - 8x^2 + 6 - 2x &= 4-2x \\
\Rightarrow 0 &= 8x^2 - 24x - 2 \\
\Rightarrow x &= \frac{3}{2} \pm \frac{\sqrt{10}}{2}.
\end{aligned}$$

$x \neq \frac{3}{2} + \frac{\sqrt{10}}{2}$ as $\operatorname{artanh}(4x+1)$ is only valid for $-1 < 4x+1 < 1 \Rightarrow -\frac{1}{2} < x < 0$. So $x = \frac{3}{2} - \frac{\sqrt{10}}{2}$ is the only real solution.

12 a)

$$y = \operatorname{arsinh} x, \qquad \text{Domain: } \{x : x \in \mathbb{R}\}, \quad \text{Range: } \{y : y \in \mathbb{R}\}.$$

12 b)

$$y = \operatorname{arcosh} x, \qquad \text{Domain: } \{x : x \in \mathbb{R},\ x \geq 1\}, \quad \text{Range: } \{y : y \in \mathbb{R},\ y \geq 0\}.$$

12 c)

$$y = \operatorname{artanh} x, \qquad \text{Domain: } \{x : x \in \mathbb{R},\ -1 < x < 1\}, \quad \text{Range: } \{y : y \in \mathbb{R}\}.$$

13 a)

$$f\left(-\frac{1}{2}\right) = \ln\left(-\frac{1}{2} + \sqrt{(-\frac{1}{2})^2 + 1}\right) = \ln\left(-\frac{1}{2} + \frac{\sqrt{5}}{2}\right),$$
$$f\left(\frac{1}{2}\right) = \ln\left(\frac{1}{2} + \sqrt{(\frac{1}{2})^2 + 1}\right) = \ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right).$$

Range: $\{f(x) : f(x) \in \mathbb{R},\ \ln(-\frac{1}{2} + \frac{\sqrt{5}}{2}) < f(x) < \ln(\frac{1}{2} + \frac{\sqrt{5}}{2})\}$.

13 b)

$$g\left(-\frac{1}{2}\right) = \frac{1}{2}\ln\left(\frac{1 + (-\frac{1}{2})}{1 - (-\frac{1}{2})}\right) = \frac{1}{2}\ln\left(\frac{1/2}{3/2}\right) = \frac{1}{2}\ln\frac{1}{3} = -\ln\sqrt{3},$$
$$g\left(\frac{1}{2}\right) = \frac{1}{2}\ln\left(\frac{1 + 1/2}{1 - 1/2}\right) = \frac{1}{2}\ln\left(\frac{3/2}{1/2}\right) = \frac{1}{2}\ln 3 = \ln\sqrt{3}.$$

Range: $\{g(x) : g(x) \in \mathbb{R},\ -\ln\sqrt{3} < g(x) < \ln\sqrt{3}\}$.

14 a)

Domain: $\{x : x \in \mathbb{R},\ x \geq 1\}$. Range: $\{y : y \in \mathbb{R},\ y \geq 1\}$.

14 b)

Domain: $\{x : x \in \mathbb{R},\ x \geq 2\}$. Range: $\{y : y \in \mathbb{R},\ y \geq 0\}$.

14 c)

Domain: $\{x : x \in \mathbb{R},\ x \geq \frac{1}{2}\}$. Range: $\{y : y \in \mathbb{R},\ y \geq 0\}$.

14 d)

Domain: $\{x : x \in \mathbb{R},\ x \geq 1\}$. Range: $\{y : y \in \mathbb{R},\ y \geq 0\}$.

14 e)

Domain: $\{x : x \in \mathbb{R},\ x \geq -1\}$. Range: $\{y : y \in \mathbb{R},\ y \geq 0\}$.

14 f)

Domain: $\{x : x \in \mathbb{R},\ x \geq 2\}$. Range: $\{y : y \in \mathbb{R},\ y \geq 0\}$.

14 g)

Domain: $\{x : x \in \mathbb{R},\ x \geq 1\}$. Range: $\{y : y \in \mathbb{R},\ y \geq -5\}$.

14 h)

Domain: $\{x : x \in \mathbb{R},\ x \leq 3\}$. Range: $\{y : y \in \mathbb{R},\ y \geq 0\}$.

14 i)

Domain: $\{x : x \in \mathbb{R},\ x \geq 1\}$. Range: $\{y : y \in \mathbb{R},\ y \leq 8\}$.

14 j)

Domain: $\{x : x \in \mathbb{R},\ x \leq 16\}$. Range: $\{y : y \in \mathbb{R},\ y \leq 0\}$.

15 a)

$$y = \operatorname{arsinh}^2 x, \qquad \text{Domain: } \{x : x \in \mathbb{R}\}, \quad \text{Range: } \{y : y \in \mathbb{R},\ y \geq 0\}.$$

15 b)

$$y = \operatorname{arcosh}^2 x, \qquad \text{Domain: } \{x : x \in \mathbb{R},\ x \geq 1\}, \quad \text{Range: } \{y : y \in \mathbb{R},\ y \geq 0\}.$$

15 c)

$$y = \operatorname{artanh}^2 x, \qquad \text{Domain: } \{x : x \in \mathbb{R},\ -1 < x < 1\}, \quad \text{Range: } \{y : y \in \mathbb{R},\ y \geq 0\}.$$

16)

$$\operatorname{arsinh} x = \ln(x + \sqrt{x^2 + 1}) \quad \text{and} \quad \operatorname{arcosh} x = \ln(x + \sqrt{x^2 - 1}).$$

Since $\sqrt{x^2 + 1} > \sqrt{x^2 - 1}$ for all $x \geq 1$, it must be that $\operatorname{arsinh} x > \operatorname{arcosh} x$ for all $x \geq 1$.

17)

$$\begin{aligned} f(1) &= \operatorname{arsinh} 1 + \operatorname{arcosh} 1 \\ &= \ln(1 + \sqrt{1^2 + 1}) + \ln(1 + \sqrt{1^2 - 1}) \\ &= \ln(1 + \sqrt{2}) + \ln 1 \\ &= \ln(1 + \sqrt{2}). \end{aligned}$$

Therefore, Range: $\{f(x) : f(x) \in \mathbb{R},\ f(x) \geq \ln(1 + \sqrt{2})\}$.

18)

$$\begin{aligned}
\operatorname{artanh} x + \ln(\sqrt{1-x}) &= \frac{1}{2}\ln 3 - 2\ln 2 \\
\Rightarrow \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right) + \ln(1-x)^{1/2} &= \frac{1}{2}\ln 3 - 2\ln 2 \\
\Rightarrow \ln\left(\frac{1+x}{1-x}\right)^{1/2} + \ln(1-x)^{1/2} &= \ln 3^{1/2} - \ln 2^2 \\
\Rightarrow \ln\left(\frac{(1+x)^{1/2}}{(1-x)^{1/2}} \times (1-x)^{1/2}\right) &= \ln\left(\frac{3^{1/2}}{4}\right) \\
\Rightarrow (1+x)^{1/2} &= \frac{3^{1/2}}{4} \\
\Rightarrow 1+x &= \frac{3}{16} \\
\Rightarrow x &= -\frac{13}{16}.
\end{aligned}$$

19)

$$\begin{aligned}
\operatorname{arsinh} x &= \ln 4 - \ln 3 + \operatorname{arcosh} x \\
\Rightarrow \operatorname{arsinh} x - \operatorname{arcosh} x &= \ln\frac{4}{3} \\
\Rightarrow \ln(x+\sqrt{x^2+1}) - \ln(x+\sqrt{x^2-1}) &= \ln\frac{4}{3} \\
\Rightarrow \ln\left(\frac{x+\sqrt{x^2+1}}{x+\sqrt{x^2-1}}\right) &= \ln\frac{4}{3} \\
\Rightarrow \frac{x+\sqrt{x^2+1}}{x+\sqrt{x^2-1}} &= \frac{4}{3} \\
\Rightarrow 3x+3\sqrt{x^2+1} &= 4x+4\sqrt{x^2-1} \\
\Rightarrow 3\sqrt{x^2+1} - 4\sqrt{x^2-1} &= x \\
\Rightarrow (3\sqrt{x^2+1} - 4\sqrt{x^2-1})^2 &= x^2 \\
\Rightarrow 9(x^2+1) - 24\sqrt{x^4-1} + 16(x^2-1) &= x^2 \\
\Rightarrow 24x^2 - 7 &= 24\sqrt{x^4-1} \\
\Rightarrow (24x^2-7)^2 &= (24\sqrt{x^4-1})^2 \\
\Rightarrow 576x^4 - 336x^2 + 49 &= 576(x^4-1) \\
\Rightarrow -336x^2 + 625 &= 0 \\
\Rightarrow x^2 &= \frac{625}{336} \\
\Rightarrow x &= \frac{25}{4\sqrt{21}} = \frac{25\sqrt{21}}{84}.
\end{aligned}$$

20)

$$f(x) = \sqrt{x^2+1} + \operatorname{arsinh} x,$$

$$f'(x) = \frac{x}{\sqrt{x^2+1}} + \frac{1}{\sqrt{x^2+1}} = \frac{x+1}{\sqrt{x^2+1}}.$$

Stationary points exist when $f'(x) = 0$:

$$\frac{x+1}{\sqrt{x^2+1}} = 0 \Rightarrow x+1 = 0 \Rightarrow x = -1.$$

$$\begin{aligned} f(-1) &= \sqrt{(-1)^2+1} + \ln\left(-1 + \sqrt{(-1)^2+1}\right) \\ &= \sqrt{2} + \ln(-1+\sqrt{2}). \end{aligned}$$

Range: $\{f(x) : f(x) \in \mathbb{R},\ f(x) \geq \sqrt{2} + \ln(-1+\sqrt{2})\}$.

21)

$$\begin{aligned} f(x) &= \sqrt{x^2-1} - 4\operatorname{arcosh} x, \\ f'(x) &= \frac{x}{\sqrt{x^2-1}} - \frac{4}{\sqrt{x^2-1}} = \frac{x-4}{\sqrt{x^2-1}}. \end{aligned}$$

Stationary points exist when $f'(x) = 0$:

$$\frac{x-4}{\sqrt{x^2-1}} = 0 \Rightarrow x-4 = 0 \Rightarrow x = 4.$$

$$\begin{aligned} f(4) &= \sqrt{4^2-1} - 4\ln(4+\sqrt{4^2-1}) \\ &= \sqrt{15} - 4\ln(4+\sqrt{15}). \end{aligned}$$

Range: $\{f(x) : f(x) \in \mathbb{R},\ f(x) \geq \sqrt{15} - 4\ln(4+\sqrt{15})\}$.

22 a)

The largest possible domain of $\operatorname{arsinh} x$ is $\{x : x \in \mathbb{R}\}$, while the largest possible domain of $\operatorname{arcosh} x$ is $\{x : x \in \mathbb{R},\ x \geq 1\}$. So the largest possible domain of $f(x) = 2\operatorname{arsinh} x - \operatorname{arcosh} x$ must be $\{x : x \in \mathbb{R},\ x \geq 1\}$ as this satisfies both of the inverse hyperbolic terms in the function.

22 b)

$$\begin{aligned} f(x) &= 2\operatorname{arsinh} x - \operatorname{arcosh} x, \\ f'(x) &= 2 \times \frac{1}{\sqrt{x^2+1}} - \frac{1}{\sqrt{x^2-1}} \\ &= \frac{2\sqrt{x^2-1} - \sqrt{x^2+1}}{\sqrt{x^2+1}\sqrt{x^2-1}}. \end{aligned}$$

Stationary points exist when $f'(x) = 0$:

$$\begin{aligned} 2\sqrt{x^2-1} - \sqrt{x^2+1} &= 0 \\ 2\sqrt{x^2-1} &= \sqrt{x^2+1} \\ 4(x^2-1) &= x^2+1 \\ 4x^2 - 4 &= x^2+1 \end{aligned}$$

$$3x^2 = 5$$
$$x^2 = \frac{5}{3}$$
$$x = \pm\sqrt{\frac{5}{3}}.$$

As $x \geq 1$, there is only one stationary point at $x = \sqrt{\frac{5}{3}}$.

22 c)

$$\begin{aligned} f\left(\sqrt{\frac{5}{3}}\right) &= 2\ln\left(\sqrt{\frac{5}{3}} + \sqrt{\left(\sqrt{\frac{5}{3}}\right)^2 + 1}\right) - \ln\left(\sqrt{\frac{5}{3}} + \sqrt{\left(\sqrt{\frac{5}{3}}\right)^2 - 1}\right) \\ &= 2\ln\left(\sqrt{\frac{5}{3}} + \frac{2\sqrt{6}}{3}\right) - \ln\left(\sqrt{\frac{5}{3}} + \frac{\sqrt{6}}{3}\right) \\ &= 2\ln\left(\frac{\sqrt{15} + 2\sqrt{6}}{3}\right) - \ln\left(\frac{\sqrt{15} + \sqrt{6}}{3}\right) \\ &= 2\ln(\sqrt{15} + 2\sqrt{6}) - \ln(\sqrt{15} + \sqrt{6}) - \ln 3. \end{aligned}$$

Range: $\{f(x) : f(x) \in \mathbb{R},\ f(x) \geq 2\ln(\sqrt{15} + 2\sqrt{6}) - \ln(\sqrt{15} + \sqrt{6}) - \ln 3\}$.

23)

$$\begin{aligned} \text{Let } y = \operatorname{cosech} x &= \frac{2}{e^x - e^{-x}}. \\ \text{Swap } x\text{s and } y\text{s: } x &= \frac{2}{e^y - e^{-y}} \\ \Rightarrow x &= \frac{2e^y}{e^{2y} - 1} \\ \Rightarrow xe^{2y} - x &= 2e^y \\ \Rightarrow xe^{2y} - 2e^y - x &= 0 \\ \Rightarrow e^{2y} - \frac{2}{x}e^y - 1 &= 0 \\ \Rightarrow \left(e^y - \frac{1}{x}\right)^2 - \frac{1}{x^2} - 1 &= 0 \\ \Rightarrow \left(e^y - \frac{1}{x}\right)^2 &= \frac{1}{x^2} + 1 \\ \Rightarrow e^y - \frac{1}{x} &= \pm\sqrt{\frac{1}{x^2} + 1} \\ \Rightarrow e^y &= \frac{1}{x} \pm \sqrt{\frac{1}{x^2} + 1}. \end{aligned}$$

We take the positive square root only, as $e^y > 0$ ($\frac{1}{x} - \sqrt{\frac{1}{x^2} + 1} < 0$ for all real values of $x \neq 0$).

$$\begin{aligned} \Rightarrow y &= \ln\left(\frac{1}{x} + \sqrt{\frac{1}{x^2} + 1}\right) \\ \therefore \operatorname{arcosech} x &= \ln\left(\frac{1}{x} + \sqrt{\frac{1}{x^2} + 1}\right). \end{aligned}$$

24)

$$\begin{aligned}
\text{Let } \frac{y}{k} = \operatorname{cosech} x &= \frac{2}{e^x - e^{-x}}. \\
\text{Swap } x\text{s and } y\text{s: } \frac{x}{k} &= \frac{2}{e^y - e^{-y}} \\
\Rightarrow \frac{x}{k} &= \frac{2e^y}{e^{2y} - 1} \\
\Rightarrow \frac{xe^{2y}}{k} - \frac{x}{k} &= 2e^y \\
\Rightarrow \frac{xe^{2y}}{k} - 2e^y - \frac{x}{k} &= 0 \\
\Rightarrow e^{2y} - \frac{2k}{x}e^y - 1 &= 0 \\
\Rightarrow \left(e^y - \frac{k}{x}\right)^2 - \frac{k^2}{x^2} - 1 &= 0 \\
\Rightarrow \left(e^y - \frac{k}{x}\right)^2 &= \frac{k^2}{x^2} + 1 \\
\Rightarrow e^y - \frac{k}{x} &= \pm\sqrt{\frac{k^2}{x^2} + 1} \\
\Rightarrow e^y &= \frac{k}{x} \pm \sqrt{\frac{k^2}{x^2} + 1}.
\end{aligned}$$

We take the positive square root only, as $e^y > 0$ ($\frac{k}{x} - \sqrt{\frac{k^2}{x^2} + 1} < 0$ for all real values of $x \neq 0$ and for $k > 0$).

$$\begin{aligned}
\Rightarrow y &= \ln\left(\frac{k}{x} + \sqrt{\frac{k^2}{x^2} + 1}\right) \\
&= \ln\left(\frac{k}{x} + \frac{1}{x}\sqrt{k^2 + x^2}\right) \\
&= \ln\left(\frac{k + \sqrt{k^2 + x^2}}{x}\right) \\
\therefore \operatorname{arcosech} \frac{x}{k} &= \ln\left(\frac{k + \sqrt{k^2 + x^2}}{x}\right).
\end{aligned}$$

25)

$$\begin{aligned}
\text{Let } y = \operatorname{sech} x &= \frac{2}{e^x + e^{-x}}. \\
\text{Swap } x\text{s and } y\text{s: } x &= \frac{2}{e^y + e^{-y}} \\
\Rightarrow x &= \frac{2e^y}{e^{2y} + 1} \\
\Rightarrow xe^{2y} + x &= 2e^y \\
\Rightarrow xe^{2y} - 2e^y + x &= 0 \\
\Rightarrow e^{2y} - \frac{2}{x}e^y + 1 &= 0
\end{aligned}$$

$$\Rightarrow \left(e^y - \frac{1}{x}\right)^2 - \frac{1}{x^2} + 1 = 0$$
$$\Rightarrow \left(e^y - \frac{1}{x}\right)^2 = \frac{1}{x^2} - 1$$
$$\Rightarrow e^y - \frac{1}{x} = \pm\sqrt{\frac{1}{x^2} - 1}$$
$$\Rightarrow e^y = \frac{1}{x} \pm \sqrt{\frac{1}{x^2} - 1}.$$

Now, $0 < x < 1$ and for $y = \operatorname{arsech} x$ the range is $y \geq 0$. Therefore, $e^y \geq 1$, so the correct root must be at least 1. For $0 < x < 1$, $\frac{1}{x} + \sqrt{\frac{1}{x^2} - 1} \geq \frac{1}{x} > 1$, so this works. For $0 < x < 1$, $\sqrt{\frac{1}{x^2} - 1} < \frac{1}{x}$ because $\frac{1}{x^2} - 1 < \frac{1}{x^2}$ and so $0 < \frac{1}{x} - \sqrt{\frac{1}{x^2} - 1} < 1$, hence the negative root cannot equal e^y.

$$\Rightarrow y = \ln\left(\frac{1}{x} + \sqrt{\frac{1}{x^2} - 1}\right)$$
$$\therefore \operatorname{arsech} x = \ln\left(\frac{1}{x} + \sqrt{\frac{1}{x^2} - 1}\right).$$

26)

$$\text{Let } \frac{y}{k} = \operatorname{sech} x = \frac{2}{e^x + e^{-x}}.$$
$$\text{Swap } x\text{s and } y\text{s: } \frac{x}{k} = \frac{2}{e^y + e^{-y}}$$
$$\Rightarrow \frac{x}{k} = \frac{2e^y}{e^{2y} + 1}$$
$$\Rightarrow \frac{xe^{2y}}{k} + \frac{x}{k} = 2e^y$$
$$\Rightarrow \frac{xe^{2y}}{k} - 2e^y + \frac{x}{k} = 0$$
$$\Rightarrow e^{2y} - \frac{2k}{x}e^y + 1 = 0$$
$$\Rightarrow \left(e^y - \frac{k}{x}\right)^2 - \frac{k^2}{x^2} + 1 = 0$$
$$\Rightarrow \left(e^y - \frac{k}{x}\right)^2 = \frac{k^2}{x^2} - 1$$
$$\Rightarrow e^y - \frac{k}{x} = \pm\sqrt{\frac{k^2}{x^2} - 1}$$
$$\Rightarrow e^y = \frac{k}{x} \pm \sqrt{\frac{k^2}{x^2} - 1}.$$

Now, $0 < \frac{x}{k} < 1$ and for $y = \operatorname{arsech} \frac{x}{k}$ the range is $y \geq 0$. Therefore, $e^y \geq 1$, so the correct root must be at least 1. For $0 < \frac{x}{k} < 1$, we have $\frac{k}{x} > 1$, so $\frac{k}{x} + \sqrt{\frac{k^2}{x^2} - 1} \geq \frac{k}{x} > 1$, so this works. For $0 < \frac{x}{k} < 1$, $\sqrt{\frac{k^2}{x^2} - 1} < \frac{k}{x}$ because $\frac{k^2}{x^2} - 1 < \frac{k^2}{x^2}$ and so $0 < \frac{k}{x} - \sqrt{\frac{k^2}{x^2} - 1} < 1$, hence the negative root cannot equal e^y.

$$\Rightarrow y = \ln\left(\frac{k}{x} + \sqrt{\frac{k^2}{x^2} - 1}\right)$$

$$
\begin{aligned}
&= \ln\left(\frac{k}{x} + \frac{1}{x}\sqrt{k^2 - x^2}\right) \\
&= \ln\left(\frac{k + \sqrt{k^2 - x^2}}{x}\right) \\
\therefore \operatorname{arsech}\frac{x}{k} &= \ln\left(\frac{k + \sqrt{k^2 - x^2}}{x}\right).
\end{aligned}
$$

27)

$$
\begin{aligned}
\text{Let } y = \coth x &= \frac{e^x + e^{-x}}{e^x - e^{-x}}. \\
\text{Swap } x\text{s and } y\text{s: } x &= \frac{e^y + e^{-y}}{e^y - e^{-y}} \\
\Rightarrow x &= \frac{e^{2y} + 1}{e^{2y} - 1} \\
\Rightarrow xe^{2y} - x &= e^{2y} + 1 \\
\Rightarrow x + 1 &= xe^{2y} - e^{2y} \\
\Rightarrow x + 1 &= e^{2y}(x - 1) \\
\Rightarrow e^{2y} &= \frac{x + 1}{x - 1} \\
\Rightarrow 2y &= \ln\left(\frac{x + 1}{x - 1}\right) \\
\Rightarrow y &= \frac{1}{2}\ln\left(\frac{x + 1}{x - 1}\right) \\
\therefore \operatorname{arcoth} x &= \frac{1}{2}\ln\left(\frac{x + 1}{x - 1}\right).
\end{aligned}
$$

28)

$$
\begin{aligned}
\text{Let } \frac{y}{k} = \coth x &= \frac{e^x + e^{-x}}{e^x - e^{-x}}. \\
\text{Swap } x\text{s and } y\text{s: } \frac{x}{k} &= \frac{e^y + e^{-y}}{e^y - e^{-y}} \\
\Rightarrow \frac{x}{k} &= \frac{e^{2y} + 1}{e^{2y} - 1} \\
\Rightarrow \frac{xe^{2y}}{k} - \frac{x}{k} &= e^{2y} + 1 \\
\Rightarrow \frac{x}{k} + 1 &= \frac{xe^{2y}}{k} - e^{2y} \\
\Rightarrow \frac{x}{k} + 1 &= e^{2y}\left(\frac{x}{k} - 1\right) \\
\Rightarrow e^{2y} &= \frac{x + k}{x - k} \\
\Rightarrow 2y &= \ln\left(\frac{x + k}{x - k}\right) \\
\Rightarrow y &= \frac{1}{2}\ln\left(\frac{x + k}{x - k}\right)
\end{aligned}
$$

$$\therefore \operatorname{arcoth}\frac{x}{k} = \frac{1}{2}\ln\left(\frac{x+k}{x-k}\right).$$

29)

$$\begin{aligned}
&\operatorname{arcosech} x = \ln 3 \\
\Rightarrow &\ln\left(\frac{1}{x} + \sqrt{\frac{1}{x^2}+1}\right) = \ln 3 \\
\Rightarrow &\frac{1}{x} + \sqrt{\frac{1}{x^2}+1} = 3 \\
\Rightarrow &\frac{1}{x} + \frac{1}{x}\sqrt{1+x^2} = 3 \\
\Rightarrow &1 + \sqrt{1+x^2} = 3x \\
\Rightarrow &\sqrt{1+x^2} = 3x-1 \\
\Rightarrow &1+x^2 = (3x-1)^2 \\
\Rightarrow &1+x^2 = 9x^2 - 6x + 1 \\
\Rightarrow &0 = 8x^2 - 6x \\
\Rightarrow &0 = 4x^2 - 3x \\
\Rightarrow &0 = x(4x-3) \\
\Rightarrow &x = 0 \text{ or } x = \frac{3}{4}.
\end{aligned}$$

$x \neq 0$ as $\operatorname{arcosech} 0$ is undefined. So the only solution is $x = \frac{3}{4}$.

Alternatively:

$$\begin{aligned}
&\operatorname{arcosech} x = \ln 3 \\
\operatorname{cosech}(\operatorname{arcosech} x) &= \operatorname{cosech}(\ln 3) \\
x &= \frac{2}{e^{\ln 3} - e^{-\ln 3}} \\
&= \frac{2}{3 - \frac{1}{3}} \\
\therefore x &= \frac{3}{4}.
\end{aligned}$$

30)

$$\begin{aligned}
&\operatorname{arsech}\frac{x}{4} = \ln 5 \\
\Rightarrow &\ln\left(\frac{4+\sqrt{16-x^2}}{x}\right) = \ln 5 \\
\Rightarrow &4 + \sqrt{16-x^2} = 5x \\
\Rightarrow &\sqrt{16-x^2} = 5x-4 \\
\Rightarrow &16-x^2 = (5x-4)^2 \\
\Rightarrow &16-x^2 = 25x^2 - 40x + 16
\end{aligned}$$

$$\begin{aligned}
&\Rightarrow 0 = 26x^2 - 40x \\
&\Rightarrow 0 = 13x^2 - 20x \\
&\Rightarrow 0 = x(13x - 20) \\
&\Rightarrow x = 0 \text{ or } x = \frac{20}{13}.
\end{aligned}$$

$x \neq 0$ as $\operatorname{arsech} 0$ is undefined. So the only solution is $x = \frac{20}{13}$.

Alternatively:

$$\begin{aligned}
\operatorname{arsech} \frac{x}{4} &= \ln 5 \\
\operatorname{sech}\left(\operatorname{arsech} \frac{x}{4}\right) &= \operatorname{sech}(\ln 5) \\
\frac{x}{4} &= \frac{2}{e^{\ln 5} + e^{-\ln 5}} \\
&= \frac{2}{5 + \frac{1}{5}} = \frac{5}{13} \\
\therefore x &= \frac{20}{13}.
\end{aligned}$$

31)

$$\begin{aligned}
&\operatorname{arcoth} x = \ln 2 \\
&\Rightarrow \frac{1}{2} \ln\left(\frac{x+1}{x-1}\right) = \ln 2 \\
&\Rightarrow \ln\left(\frac{x+1}{x-1}\right) = 2 \ln 2 \\
&\Rightarrow \ln\left(\frac{x+1}{x-1}\right) = \ln 4 \\
&\Rightarrow \frac{x+1}{x-1} = 4 \\
&\Rightarrow x + 1 = 4x - 4 \\
&\Rightarrow 3x = 5 \\
&\Rightarrow x = \frac{5}{3}.
\end{aligned}$$

Alternatively:

$$\begin{aligned}
\operatorname{arcoth} x &= \ln 2 \\
\coth(\operatorname{arcoth} x) &= \coth(\ln 2) \\
x &= \frac{e^{\ln 2} + e^{-\ln 2}}{e^{\ln 2} - e^{-\ln 2}} \\
&= \frac{2 + \frac{1}{2}}{2 - \frac{1}{2}} \\
\therefore x &= \frac{5}{3}.
\end{aligned}$$

32 a)

$$\begin{aligned}
\operatorname{arcosech} x &= \ln\left(2+\frac{1}{x}\right) \\
\Rightarrow \ln\left(\frac{1}{x}+\sqrt{\frac{1}{x^2}+1}\right) &= \ln\left(2+\frac{1}{x}\right) \\
\Rightarrow \sqrt{\frac{1}{x^2}+1} &= 2 \\
\Rightarrow \frac{1}{x^2}+1 &= 4 \\
\Rightarrow \frac{1}{x^2} &= 3 \\
\Rightarrow x^2 &= \frac{1}{3} \\
\Rightarrow x &= \pm\frac{\sqrt{3}}{3}.
\end{aligned}$$

Alternatively:

$$\begin{aligned}
\operatorname{arcosech} x &= \ln\left(2+\frac{1}{x}\right) \\
\operatorname{cosech}(\operatorname{arcosech} x) &= \operatorname{cosech}\left(\ln\left(2+\frac{1}{x}\right)\right) \\
x &= \frac{2}{(2+\frac{1}{x})-\frac{1}{2+\frac{1}{x}}} \\
&= \frac{2}{2+\frac{1}{x}-\frac{x}{2x+1}} \\
\Rightarrow x\left(2+\frac{1}{x}-\frac{x}{2x+1}\right) &= 2 \\
\Rightarrow 2x+1-\frac{x^2}{2x+1} &= 2 \\
\Rightarrow (2x+1)^2-x^2-2(2x+1) &= 0 \\
\Rightarrow 4x^2+4x+1-x^2-4x-2 &= 0 \\
\Rightarrow 3x^2 &= 1 \\
\Rightarrow x^2 &= \frac{1}{3} \\
\Rightarrow x &= \pm\frac{1}{\sqrt{3}} \\
\therefore x &= \pm\frac{\sqrt{3}}{3}.
\end{aligned}$$

32 b)

$$\begin{aligned}
\operatorname{arsech}\frac{x}{2} &= \ln\left(3-\frac{4}{x}\right) \\
\Rightarrow \ln\left(\frac{2+\sqrt{4-x^2}}{x}\right) &= \ln\left(3-\frac{4}{x}\right) \\
\Rightarrow 2+\sqrt{4-x^2} &= 3x-4
\end{aligned}$$

$$\begin{aligned}
\Rightarrow \sqrt{4-x^2} &= 3x-6 \\
\Rightarrow 4-x^2 &= (3x-6)^2 \\
\Rightarrow 4-x^2 &= 9x^2-36x+36 \\
\Rightarrow 0 &= 10x^2-36x+32 \\
\Rightarrow 0 &= 5x^2-18x+16 \\
\Rightarrow x &= \frac{8}{5} \text{ or } x=2.
\end{aligned}$$

$x \neq \frac{8}{5}$ as $\ln\left(\frac{2+\sqrt{4-(\frac{8}{5})^2}}{\frac{8}{5}}\right) = \ln 2$ but $\ln\left(3-\frac{4}{\frac{8}{5}}\right) = \ln\frac{1}{2}$ and $\ln 2 \neq \ln\frac{1}{2}$. So the only real solution is $x=2$.

Alternatively:

$$\begin{aligned}
\operatorname{arsech}\frac{x}{2} &= \ln\left(3-\frac{4}{x}\right) \\
\operatorname{sech}\left(\operatorname{arsech}\frac{x}{2}\right) &= \operatorname{sech}\left(\ln\left(3-\frac{4}{x}\right)\right) \\
\frac{x}{2} &= \frac{2}{3-\frac{4}{x}+\frac{1}{3-\frac{4}{x}}} \\
x &= \frac{4}{3-\frac{4}{x}+\frac{x}{3x-4}} \\
\Rightarrow x\left(3-\frac{4}{x}+\frac{x}{3x-4}\right) &= 4 \\
\Rightarrow 3x-4+\frac{x^2}{3x-4} &= 4 \\
\Rightarrow (3x-4)^2+x^2-4(3x-4) &= 0 \\
\Rightarrow 9x^2-24x+16+x^2-12x+16 &= 0 \\
\Rightarrow 10x^2-36x+32 &= 0 \\
\Rightarrow 5x^2-18x+16 &= 0 \\
\Rightarrow x &= \frac{8}{5} \text{ or } x=2.
\end{aligned}$$

$x \neq \frac{8}{5}$ as $\ln\left(\frac{2+\sqrt{4-(\frac{8}{5})^2}}{\frac{8}{5}}\right) = \ln 2$ but $\ln\left(3-\frac{4}{\frac{8}{5}}\right) = \ln\frac{1}{2}$ and $\ln 2 \neq \ln\frac{1}{2}$. So the only real solution is $x=2$.

32 c)

$$\begin{aligned}
\operatorname{arcoth} 3x &= \ln\left(\sqrt{3+4x}\right) \\
\Rightarrow \frac{1}{2}\ln\left(\frac{3x+1}{3x-1}\right) &= \ln\left(\sqrt{3+4x}\right) \\
\Rightarrow \ln\left(\frac{3x+1}{3x-1}\right) &= 2\ln\left(\sqrt{3+4x}\right) \\
\Rightarrow \ln\left(\frac{3x+1}{3x-1}\right) &= \ln\left((\sqrt{3+4x})^2\right) \\
\Rightarrow \frac{3x+1}{3x-1} &= 3+4x \\
\Rightarrow 3x+1 &= (3+4x)(3x-1)
\end{aligned}$$

$$\Rightarrow 3x+1 = 12x^2+5x-3$$
$$\Rightarrow 0 = 12x^2+2x-4$$
$$\Rightarrow 0 = 6x^2+x-2$$
$$\Rightarrow x = -\frac{2}{3} \text{ or } x = \frac{1}{2}.$$

Alternatively:

$$\operatorname{arcoth} 3x = \ln\left(\sqrt{3+4x}\right)$$
$$\coth(\operatorname{arcoth} 3x) = \coth(\ln\left(\sqrt{3+4x}\right))$$
$$3x = \frac{\sqrt{3+4x}+\frac{1}{\sqrt{3+4x}}}{\sqrt{3+4x}-\frac{1}{\sqrt{3+4x}}}$$
$$= \frac{3+4x+1}{3+4x-1}$$
$$= \frac{4+4x}{2+4x}$$
$$\Rightarrow 3x(2+4x) = 4+4x$$
$$\Rightarrow 6x+12x^2-4-4x = 0$$
$$\Rightarrow 12x^2+2x-4 = 0$$
$$\Rightarrow 6x^2+x-2 = 0$$
$$\Rightarrow x = -\frac{2}{3} \text{ or } x = \frac{1}{2}.$$

33)

$$\operatorname{artanh}\frac{x}{2} = \operatorname{arcoth}(3x+1)$$
$$\Rightarrow \frac{1}{2}\ln\left(\frac{2+x}{2-x}\right) = \frac{1}{2}\ln\left(\frac{(3x+1)+1}{(3x+1)-1}\right)$$
$$\Rightarrow \ln\left(\frac{2+x}{2-x}\right) = \ln\left(\frac{3x+2}{3x}\right)$$
$$\Rightarrow \frac{2+x}{2-x} = \frac{3x+2}{3x}$$
$$\Rightarrow 3x(2+x) = (3x+2)(2-x)$$
$$\Rightarrow 6x+3x^2 = -3x^2+4x+4$$
$$\Rightarrow 6x^2+2x-4 = 0$$
$$\Rightarrow 3x^2+x-2 = 0$$
$$\Rightarrow x = -1 \text{ or } x = \frac{2}{3}.$$